



HACETTEPE UNIVERSITY

Faculty of Engineering

Department of Nuclear Energy Engineering

Project 2

Flow Analysis in a PWR Fuel Rod Subchannel

NEM 358 – Computational Fluid Dynamics Applications

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Contents

1	Introduction	2
2	Methodology	2
2.1	Computational Domain and Geometry	2
2.2	Mesh Generation	3
2.3	Governing Equations	3
2.4	Material Properties	3
2.5	Boundary Conditions	4
2.6	Solver Settings	4
3	Results and Discussion	4
3.1	Solution Stability and Convergence Criteria	4
3.2	Pressure Distribution at the Inlet Surface	5
3.3	Axial Pressure Profile	6
3.4	Total Pressure Drop and Analytical Comparison	7
3.5	Velocity Contour and Vector Distribution at the Outlet	8
3.6	Axial Velocity Profile and Entrance Region	9
4	Conclusion	9
A	Step-by-Step Analytical Pressure Drop	11
B	Plant-Level Input Parameters	12

1 Introduction

The PWR reactor is currently the most widely used and popular type of nuclear reactor in the world. For both PWRs and other reactor types, thermohydraulic design is extremely important to maintain the structural integrity of the fuel elements and optimize heat transfer efficiency. In the case of a PWR, the coolant flows through narrow passages between four adjacent rods; this region is called the bottom channel. These bottom channels are the regions where the flow reaches its highest turbulent intensity and heat transfer is maximized.

The aim of this project report is to investigate the flow behavior in a 3.66-meter-long bare-rod bottom channel geometry designed for a typical 4-cycle PWR reactor. In general, for straight circular tubes or rods, the pressure drop can be calculated semi-analytically using a classical approach with correlations for fully turbulent pipe flow such as the Darcy–Weisbach equation and the Filonenko equation. However, as with PWRs, the actual subchannel geometry differs from flow in a straight circular cross-section pipe due to the emergence of low-velocity regions at the concave corners formed by the rod surfaces and the complete development of flow after a certain inlet length. Therefore, not only analytical solutions but also numerical methods are necessary to better understand the velocity and pressure distribution.

Accordingly, solutions were obtained using ANSYS Fluent and Siemens STAR-CCM+, two different computational fluid dynamics (CFD) software programs accepted in the industry. The numerical results obtained from the analyses were compared both with each other and with the analytical calculation performed using the Darcy–Weisbach approach. For the analytical solution, the thermophysical properties of the coolant, i.e., water, were determined according to the IAPWS-IF97 standards under the conditions of $T = 309.4^\circ\text{C}$ and 15.51 MPa pressure. In summary, this project allowed us to observe how close the results given by two different software programs are to each other and how close they approach the analytical result in the same problem.

2 Methodology

2.1 Computational Domain and Geometry

A single subchannel of a square-lattice fuel bundle, with rod diameter $D = 9.5\text{ mm}$ and rod pitch $P = 12.6\text{ mm}$, extruded over the active fuel height $H = 3.66\text{ m}$. No spacer grids

are included.

$$A = P^2 - \frac{\pi D^2}{4} = 8.788 \times 10^{-5} \text{ m}^2 \quad (1)$$

$$P_w = \pi D = 0.02985 \text{ m} \quad (2)$$

$$D_h = \frac{4A}{P_w} = 11.78 \text{ mm} \quad (3)$$

2.2 Mesh Generation

- **STAR-CCM+:** surface remesher + trimmed cell mesher + prism layer mesher; axial trimmer stretching of 2000 %. Final cell count $\approx 920,000$.
- **ANSYS Fluent:** sweep method with 1000 axial divisions; 5-layer inflation on rod walls (first layer 0.3 mm, growth ratio 1.2). Final cell count $\approx 873,000$.

2.3 Governing Equations

The steady RANS equations are solved under the incompressible-flow assumption with two different turbulence closures: realizable k - ε in STAR-CCM+ and k - ω SST in ANSYS Fluent. Gravity acts along $-z$ with $g = 9.81 \text{ m/s}^2$.

The frictional pressure drop is given by the Darcy–Weisbach equation:

$$\Delta P_{\text{friction}} = f \cdot \frac{L}{D_h} \cdot \frac{\rho V^2}{2} \quad (4)$$

The friction factor is from the Filonenko correlation:

$$f = \left(0.790 \ln(\text{Re}) - 1.64\right)^{-2} \quad (5)$$

The Reynolds number is built on the hydraulic diameter:

$$\text{Re} = \frac{\rho V D_h}{\mu} \quad (6)$$

The hydrostatic component over a vertical channel is $\Delta P_{\text{hydro}} = \rho g H$, and the total drop is

$$\Delta P_{\text{total}} = \Delta P_{\text{friction}} + \Delta P_{\text{hydro}}. \quad (7)$$

Step-by-step evaluation is given in Appendix A.

2.4 Material Properties

CFD inputs: $\rho = 734.5 \text{ kg/m}^3$, $\mu = 9.0 \times 10^{-5} \text{ Pa s}$, isothermal, operating reference 15.51 MPa.

Analytical reference (IAPWS-IF97 at the bulk-mean temperature):

$$T_{\text{avg}} = T_{\text{in}} + \frac{1}{2}\Delta T = 292.7 + \frac{33.4}{2} = 309.4\text{ }^{\circ}\text{C} \quad (8)$$

$$\rho_{\text{avg}} = 706.16\text{ kg/m}^3, \quad \mu_{\text{avg}} = 8.48 \times 10^{-5}\text{ Pa s}$$

2.5 Boundary Conditions

- **Inlet:** $G = 3729\text{ kg/m}^2\text{ s}$ ($\dot{m} = 0.328\text{ kg/s}$ in STAR-CCM+, $V = 5.077\text{ m/s}$ in ANSYS Fluent), turbulence intensity 5 %, $D_h = 11.78\text{ mm}$.
- **Outlet:** pressure outlet, $P_{\text{gauge}} = 0\text{ Pa}$.
- **Rod walls:** no-slip stationary walls.
- **Side faces:** symmetry planes.

2.6 Solver Settings

STAR-CCM+: segregated flow solver, second-order upwind. ANSYS Fluent: coupled pressure-velocity solver, second-order upwind, pseudo-transient option enabled. Convergence is declared at residuals of order 10^{-4} , with the inlet/outlet mass balance used as an independent check.

3 Results and Discussion

3.1 Solution Stability and Convergence Criteria

The accuracy, or in other words the physical consistency, of numerical analyses is directly related to the level of convergence of the conservation equations (mass, momentum, and turbulence).

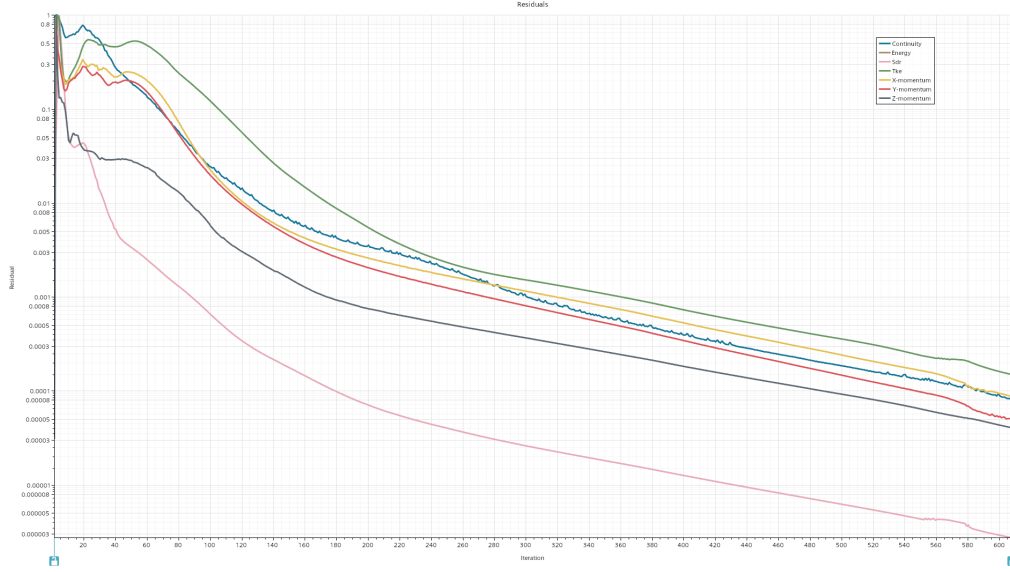


Figure 1: Residual history of the STAR-CCM+ solution.

For the numerical solution to be meaningful, the residual values must decrease sufficiently in each iteration. The residual graph in Figure 1 shows the results of the analysis performed with STAR-CCM+. In this simulation, when the iteration-dependent change of the residual values is examined, it is seen that the solution reaches full stability after approximately 600 iterations.

While the residual values for continuity and momentum (x, y, z) fall below 10^{-4} , the energy and specific dissipation ratio (Sdr) values fall to the order of 10^{-6} ; this proves that the numerical model provides mass and energy balance with high accuracy. These levels are acceptable for this project. The fluctuations seen in the graph during the first 100 iterations indicate the initial transition phase where the initial conditions in the flow field become consistent with turbulence models.

In the solution using ANSYS Fluent, since the $k-\omega$ SST model and the combined pressure-velocity solver were used, autoconvergence was achieved in approximately 174 iterations, reaching the same acceptable residual value level. This demonstrates that both solvers can numerically solve the problem stably.

3.2 Pressure Distribution at the Inlet Surface

The pressure contour graph obtained at the inlet surface reflects the symmetry of the lower channel and the wall effects. The pressure is lowest at the geometric center of the channel; the velocity decreases as it approaches the rod surfaces, causing the static pressure to rise slightly. This behavior is a direct consequence of Bernoulli's principle, which almost everyone has heard of at least once: static pressure decreases in the central region where the velocity is high and is relatively higher in the region near the walls where the velocity is low. The pressure range at the inlet section is approximately 100 Pa. Consequently,

when the contour image (Figure 2) is examined, it is clearly understood that the flow has a homogeneous distribution at the moment of entry into the channel. The pressure distribution obtained on the same surface in ANSYS Fluent also exhibited a qualitatively similar pattern, and although the color scale distribution is slightly different due to the different mesh structure, it can be said that it has a similar homogeneous distribution.

Simcenter STAR-CCM+

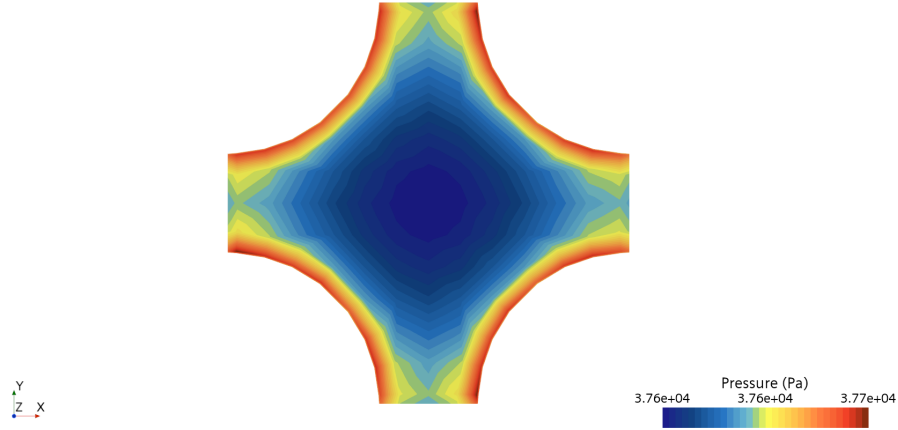


Figure 2: Static pressure contour at the inlet surface (STAR-CCM+).

3.3 Axial Pressure Profile

The axial pressure profile obtained along the channel's centerline (Figure 3) clearly shows how the flow undergoes pressure loss along its length. The curve obtained in both software programs exhibits an almost completely linear decrease from inlet to outlet. This behavior indicates that the flow enters a fully enhanced turbulent regime along the channel; since the pressure gradient per unit length is constant in a fully enhanced flow, the axial profile is a straight line. In the STAR-CCM+ profile, a distribution starting at approximately 37.5 kPa along the centerline and decreasing to zero at the outlet is observed, while in the ANSYS Fluent profile, a linear decrease starting at approximately 40.7 kPa and decreasing to zero is observed. This difference between the two software programs is a deviation within normal limits that may be due to differences in both turbulence models ($k-\epsilon$ Realizable / $k-\omega$ SST) and mesh resolution. Importantly, both curves show the same qualitative behavior, namely a linear decrease with a constant slope, which is crucial for the accuracy of the solution. Additionally, these graphs do not include the pressure difference caused by gravity.

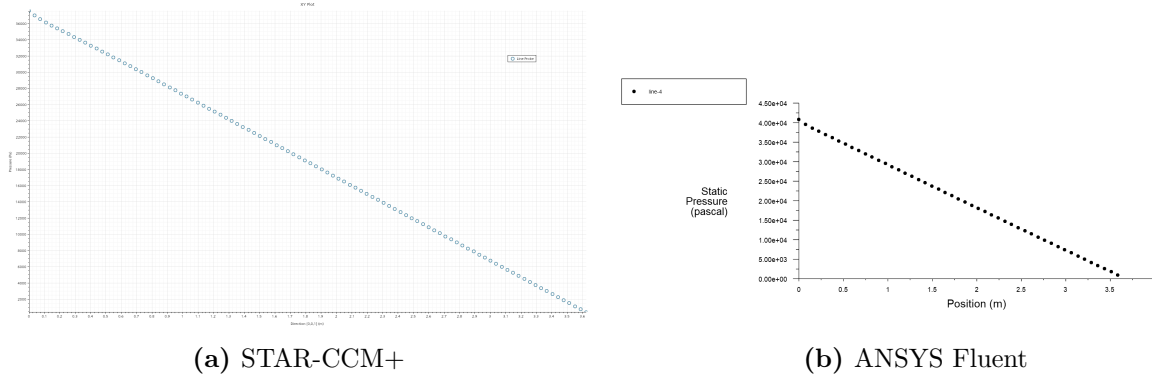


Figure 3: Axial static pressure profile along the centerline of the subchannel.

3.4 Total Pressure Drop and Analytical Comparison

The total pressure loss experienced by the fluid along the channel is the sum of two components: frictional loss and hydrostatic (gravity) loss. The frictional component was calculated using the Darcy–Weisbach formula given in Eq. (4), with the Filonenko correlation (Eq. (5)) supplying the friction factor.

The Reynolds number obtained from the calculations was approximately 5.18×10^5 , indicating that the flow is in a highly turbulent regime. Using the Filonenko correlation, the total pressure drop was found to be 65.28 kPa. This value is the analytical solution. In the STAR-CCM+ solution, the total pressure difference is 63.59 kPa, while in ANSYS Fluent, by adding the hydrostatic component to the static ΔP , a value of 67.05 kPa was obtained. The deviation between the analytical value and STAR-CCM+ is approximately 2.59 %, while the deviation between the analytical value and ANSYS Fluent is approximately 2.71 %. Both models remain below the critical error limit of 3 %, and therefore these differences can be considered within acceptable engineering limits.

The main reasons for the inconsistencies include the fact that analytical calculations assume the flow as fully developed flow from start to finish, while CFD physically captures the additional pressure drop in the inlet region; another difference is that turbulence models produce small variations in their solutions in regions near the walls. To give a direct example from this project, the higher pressure drop estimate of ANSYS stems from the fact that the $k-\omega$ SST turbulence model tends to calculate wall shear stresses higher than the $k-\varepsilon$ model. A side-by-side comparison of all three approaches is given in Table 1.

Table 1: Comparison of total pressure drop results.

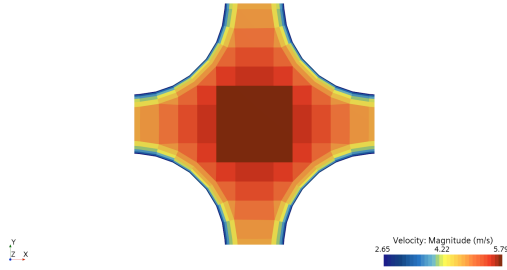
Method	ΔP (kPa)	Deviation
Analytical (mean density, IAPWS-IF97)	65.28	—
STAR-CCM+ ($k-\varepsilon$ Realizable)	63.59	2.59 %
ANSYS Fluent ($k-\omega$ SST)	67.05	2.71 %

3.5 Velocity Contour and Vector Distribution at the Outlet

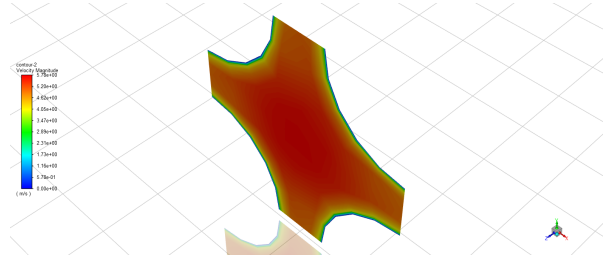
The velocity contours obtained at the exit surface (Figure 4) clearly demonstrate the typical characteristics of the subchannel velocity profile. The central region represents the highest velocity values, while the regions adjacent to the rod walls represent the lowest velocity values. In the STAR-CCM+ solution, the maximum velocity is approximately 5.79 m/s, decreasing to approximately 2.65 m/s in regions near the wall. Due to the no-slip condition, the velocity at the wall is theoretically zero; the measured value of 2.65 m/s actually reflects not the average surface velocity of the wall, but the magnitude of the velocity in the first layer of the mesh cells. This is important because the velocity does not suddenly drop to zero, but decreases gradually.

The contour obtained in ANSYS Fluent also shows qualitatively the same behavior; a maximum of approximately 5.78 m/s is observed in the center, and a decreasing slope towards zero is seen in regions near the wall. It can be said that both software programs accurately capture the same physical mechanism, namely viscous braking at the wall and the occurrence of the maximum velocity in the center. Additionally, analytical calculations revealed an average axial velocity of 5.281 m/s.

Simcenter STAR-CCM+



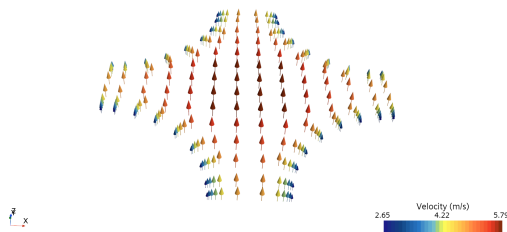
(a) STAR-CCM+



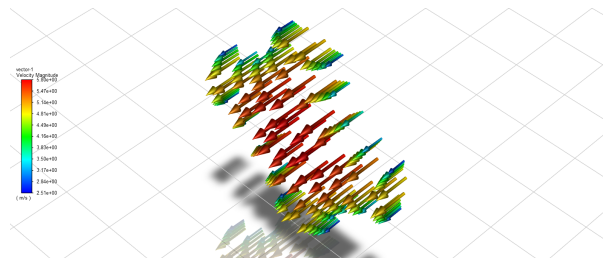
(b) ANSYS Fluent

Figure 4: Velocity magnitude contour at the outlet surface.

Simcenter STAR-CCM+



(a) STAR-CCM+



(b) ANSYS Fluent

Figure 5: Velocity vector distribution at the outlet surface.

The vector graphics in Figure 5 show that, as expected in bare-bar geometry, flow

occurs only in the main flow direction (z-axis) and no secondary flow or vortex formation occurs due to the absence of obstructing structures such as intermediate grids. In other words, this confirms that turbulence is homogeneously distributed within the channel.

3.6 Axial Velocity Profile and Entrance Region

When examining the velocity profile graphs obtained along the channel center (Figure 6), it is observed that the velocity increases abruptly from 5.08 m/s to 5.72–5.79 m/s within the first 0.4 m from the entrance.

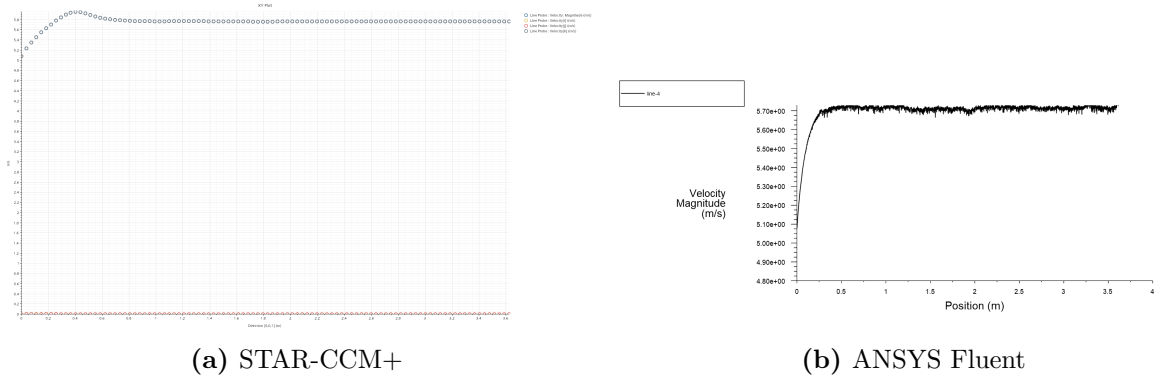


Figure 6: Axial velocity magnitude profile along the centerline of the subchannel.

This region represents the hydraulic entrance length; here the boundary layers develop at the edges narrow towards the channel center, increasing the central velocity according to the continuity equation. The stabilization of the velocity after 0.5 m proves that the fluid has passed the inlet boundary condition and entered the fully developed flow region where the viscous boundary layer merges and the velocity profile no longer changes, approximately 3.1–3.2 m. This constancy of axial velocity is fully consistent with the analytically predicted conservation of mass.

4 Conclusion

In this study, a typical bare rod subchannel of a 4-loop PWR core was modeled under steady-state, isothermal, and gravity-inclusive conditions using both STAR-CCM+ and ANSYS Fluent. Results from both software programs were solved analytically, both individually and using the classical Darcy–Weisbach approach, then compared and analyzed.

The results in terms of total pressure drop are as follows: analytical value calculated from average density along the channel: 65.28 kPa; STAR-CCM+ solution: 63.59 kPa; ANSYS Fluent solution: 67.05 kPa. Both simulations showed a deviation of less than 3 % from the analytical value. This indicates that both the k – ε Realizable and k – ω SST models, as well as STAR-CCM+ and ANSYS Fluent, yielded similar results, demonstrating that

both turbulence approaches and CFD applications can be reliably used at the subchannel scale in such flows.

In terms of velocity profile, both software programs captured the physically expected behavior. The velocity, which was smooth at the inlet, became fully developed after an inlet length of approximately 0.35–0.4 m and stabilized at approximately 5.75 m/s along the centerline. In the outlet section, the velocity contours showed maximum velocity values at the center (5.79 m/s) and minimum values near the wall, demonstrating that the viscous wall effect and the no-slip condition were correctly calculated.

The alignment of all arrows in the axial direction in the vector graphs and the negligibility of the transverse components not only confirmed the unidirectional and fully developed nature of the flow but also proved the absence of a secondary flow or vortex.

In conclusion, modern CFD tools can successfully perform thermohydraulic analyses at the PWR subchannel scale with classical correlations with an error rate of less than 3 %. This level of accuracy is within the accepted limits for engineering design and safety analysis. However, some points still need to be considered. The small discrepancies between the two applications and the direct impact of mesh resolution and turbulence model selection on the result highlight the need for extreme precision for real-world applications.

Finally, this project and these results demonstrate that CFD models are a reliable design tool for basic bare-bar geometries without interstitial grids. In later stages, different conditions and structures, such as heat flow and interstitial grids, can be added to the model; this creates a critical and important infrastructure for much more accurate modeling of real operating conditions in the reactor core.

References

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A Step-by-Step Analytical Pressure Drop

A.1 Geometry

$$A = P^2 - \frac{\pi D^2}{4} = (0.0126)^2 - \frac{\pi(0.0095)^2}{4} = 8.788 \times 10^{-5} \text{ m}^2 \quad (9)$$

$$P_w = \pi D = \pi(0.0095) = 0.02985 \text{ m} \quad (10)$$

$$D_h = \frac{4A}{P_w} = \frac{4 \times 8.788 \times 10^{-5}}{0.02985} = 0.01178 \text{ m} \quad (11)$$

A.2 Coolant Properties at Mean Bulk Temperature

$$T_{\text{avg}} = T_{\text{in}} + \frac{1}{2}\Delta T = 292.7 + \frac{33.4}{2} = 309.4 \text{ }^\circ\text{C}$$

At 309.4 °C and 15.51 MPa (IAPWS-IF97):

$$\rho_{\text{avg}} = 706.16 \text{ kg/m}^3, \quad \mu_{\text{avg}} = 8.48 \times 10^{-5} \text{ Pa s}$$

A.3 Mean Axial Velocity and Reynolds Number

$$V = \frac{G}{\rho_{\text{avg}}} = \frac{3729}{706.16} = 5.281 \text{ m/s} \quad (12)$$

$$\text{Re} = \frac{\rho_{\text{avg}} V D_h}{\mu_{\text{avg}}} = \frac{706.16 \times 5.281 \times 0.01178}{8.48 \times 10^{-5}} \approx 5.18 \times 10^5 \quad (13)$$

A.4 Filonenko Friction Factor

$$f = \left(0.790 \ln(5.18 \times 10^5) - 1.64\right)^{-2} = 0.01305 \quad (14)$$

A.5 Frictional Pressure Drop

$$\Delta P_{\text{friction}} = 0.01305 \times \frac{3.66}{0.01178} \times \frac{706.16 \times (5.281)^2}{2} \quad (15)$$

$$\approx 39\,921 \text{ Pa} = 39.92 \text{ kPa} \quad (16)$$

A.6 Hydrostatic Pressure Drop

$$\Delta P_{\text{hydro}} = \rho_{\text{avg}} g H = 706.16 \times 9.81 \times 3.66 \approx 25\,354 \text{ Pa} = 25.35 \text{ kPa} \quad (17)$$

A.7 Total Analytical Pressure Drop

$$\Delta P_{\text{total}} = \Delta P_{\text{friction}} + \Delta P_{\text{hydro}} = 39.92 + 25.35 = 65.28 \text{ kPa} \quad (18)$$

B Plant-Level Input Parameters

Table 2: Selected design parameters of a typical 4-loop PWR used as input for the analysis.

Parameter	Value
System pressure	15.51 MPa
Core inlet temperature	292.7 °C
Average temperature rise in reactor	33.4 °C
Average core inlet mass flux G	3729 kg/m ² s
Fuel rod outer diameter D	9.5 mm
Fuel rod pitch P	12.6 mm
Active fuel height H	3.66 m